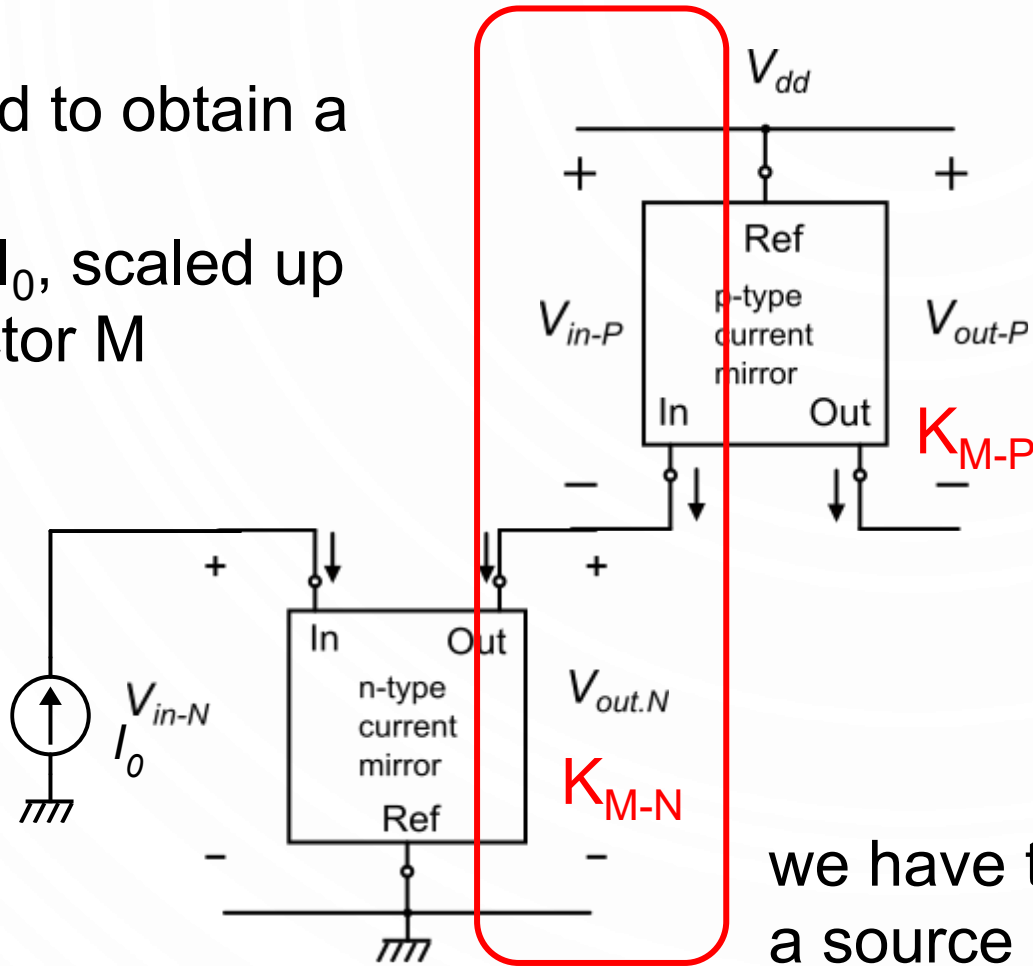


Current Mirrors

ROLE OF V_{IN} , V_{MIN} : AN EXAMPLE

We need to obtain a copy of current I_0 , scaled up by a factor M



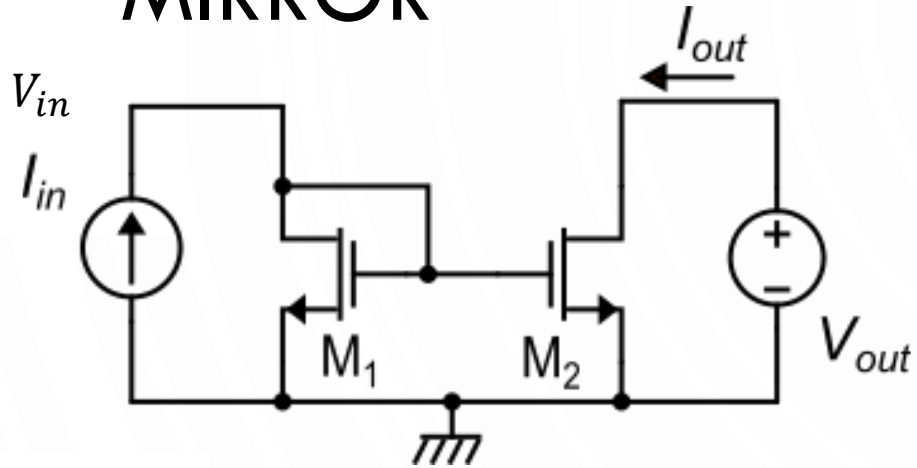
We add a p-type mirror to reverse the current source type

$$M = k_{M-N}k_{M-P}$$

$$V_{dd} = V_{out-N} + V_{in-P}$$

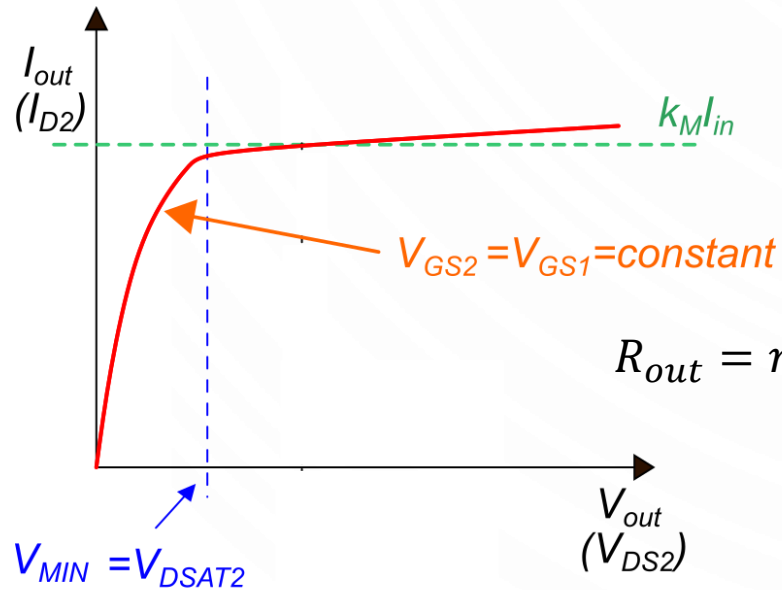
we have transformed a source into a sink

PARAMETERS OF THE SIMPLE MOSFET CURRENT MIRROR



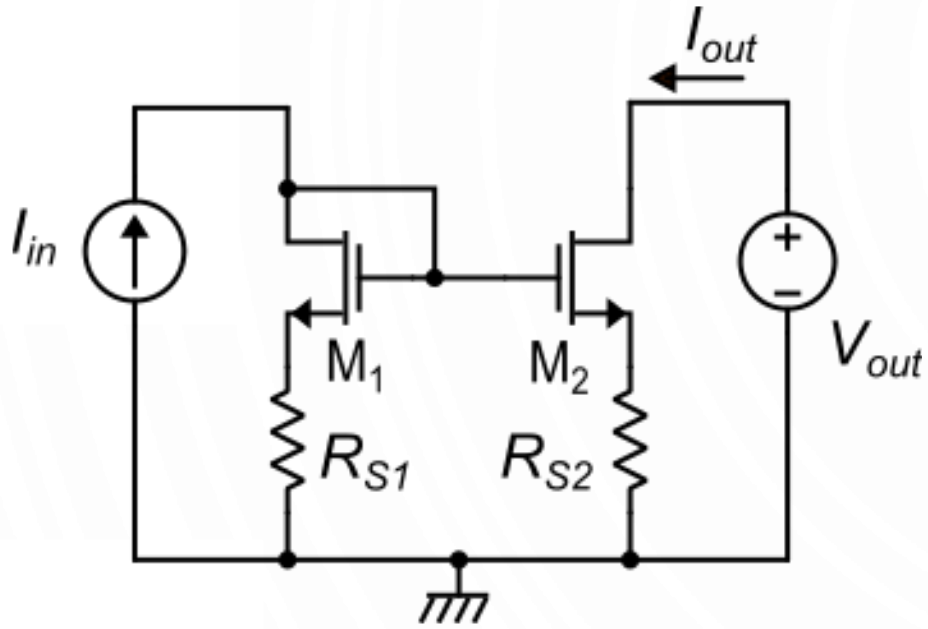
$$V_{in} = V_{GS1} = V_{GS2}$$

$$V_{MIN} = V_{DSAT2}$$

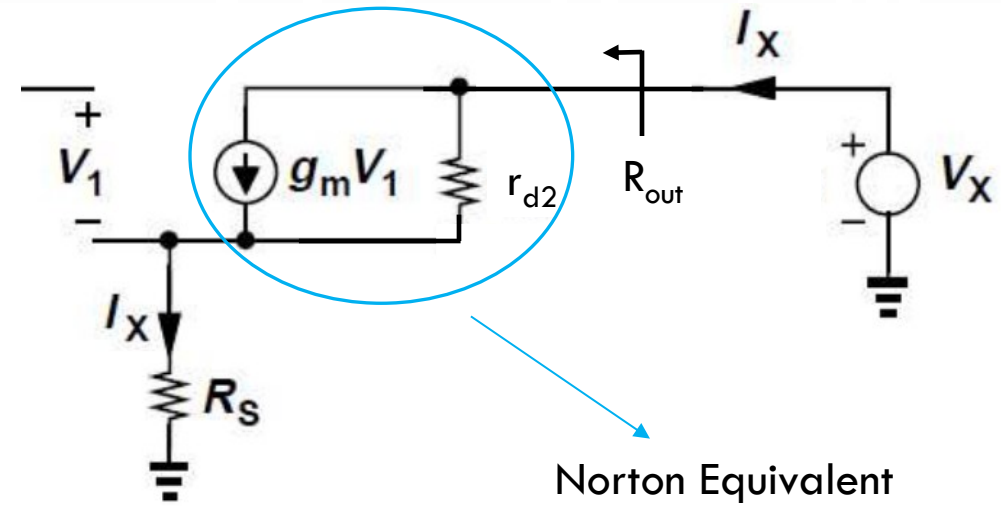


$$R_{out} = r_{d2} = \frac{1}{\lambda_2 I_{out}}$$

INCREASING R_{OUT} : SOURCE DEGENERATION



$$R_{out} = R_{S2} + r_{d2}(1 + g_{m2}R_{S2})$$

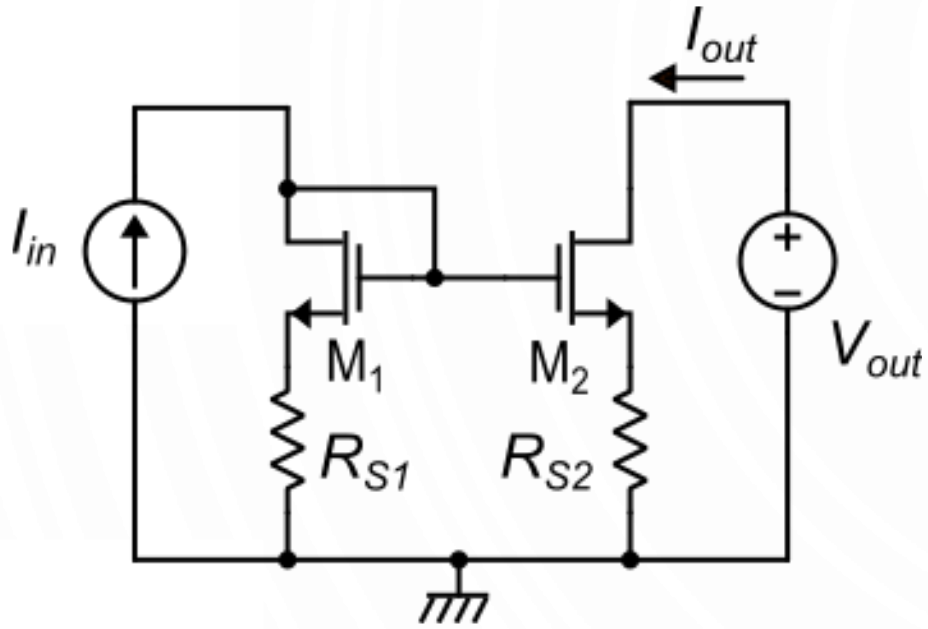


Transforming to Thevenin:

$$R_{out} = \frac{I_x R_S + g_m I_x R_S r_{d2} + I_x r_{d2}}{I_x}$$

$$R_{out} = R_S + g_m R_S r_{d2} + r_{d2} = R_S + r_{d2}(1 + g_m R_S)$$

INCREASING R_{OUT} : SOURCE DEGENERATION



$$R_{out} = R_{S2} + r_{d2}(1 + g_{m2}R_{S2})$$

$$R_{out} \cong r_{d2}(1 + g_{m2}R_{S2})$$

The output resistance is increased by a factor $(1 + g_{m2}R_{S2})$ with respect to the simple current mirror

In order to have a linear behavior, so that $I_{out} = k_M I_{in}$, we need that:

Notice:

$$V_{S2} = V_{S1} \Rightarrow R_{S1}I_{in} = R_{S2}I_{out}$$



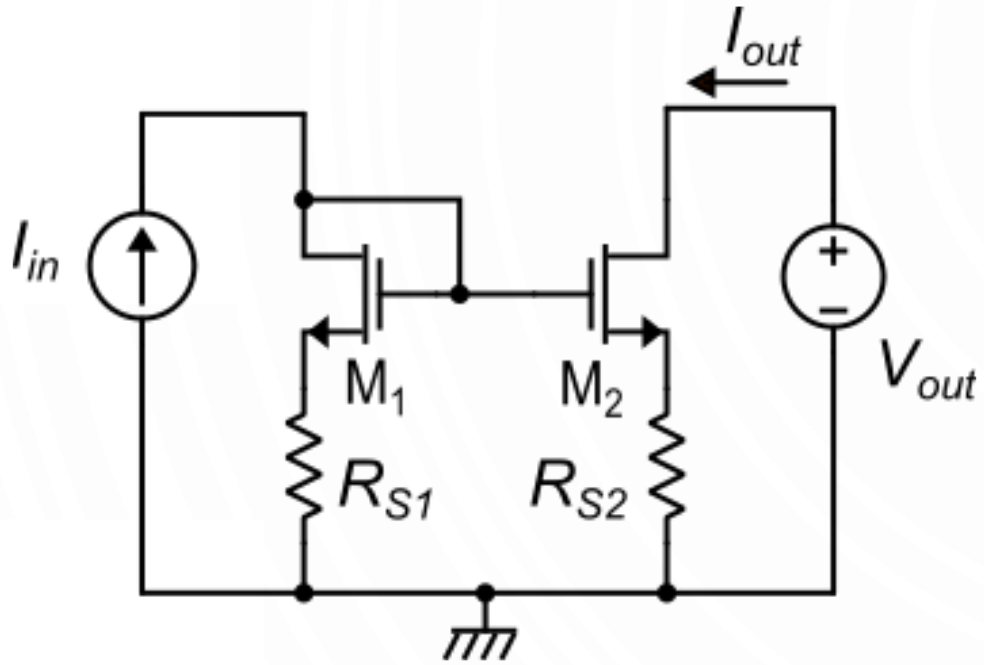
$$V_{GS1} = V_{GS2} \Rightarrow V_{S2} = V_{S1}$$

design rule

$$\frac{R_{S1}}{R_{S2}} = \frac{I_{out}}{I_{in}} = k_M$$



INCREASING R_{OUT} : SOURCE DEGENERATION



$$V_{out} = V_{DS2} + R_{S2}I_{out}$$

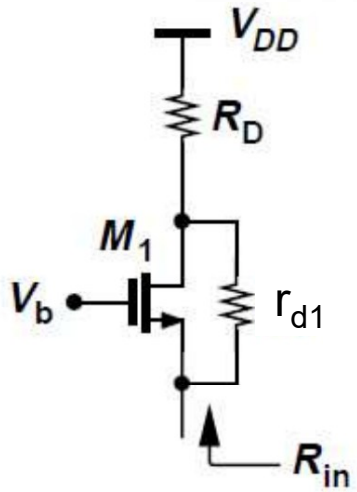
The mirror starts to fail at a V_{out} value that make M_2 enter triode region

$$V_{in} = V_{GS1} + R_{S1}I_{in}$$

Notice: $R_{out} \cong r_{d2}(1 + g_{m2}R_{S2})$

$$V_{MIN} = V_{DSAT2} + R_{S2}I_{out}$$

THE COMMON GATE STAGE



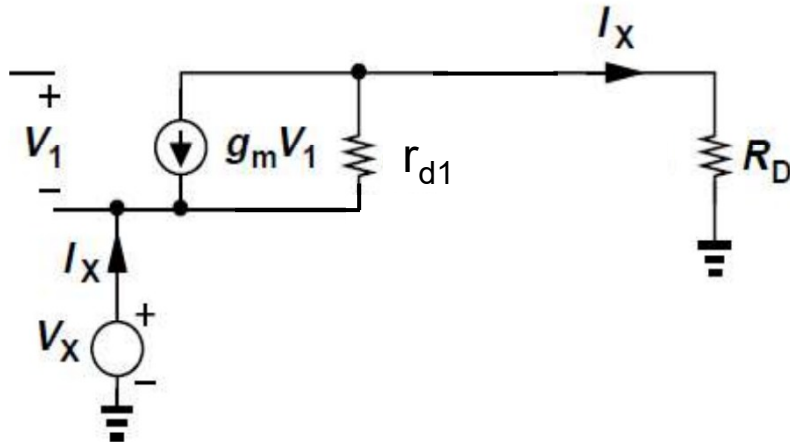
Input Impedance:

$$V_1 = -V_x$$

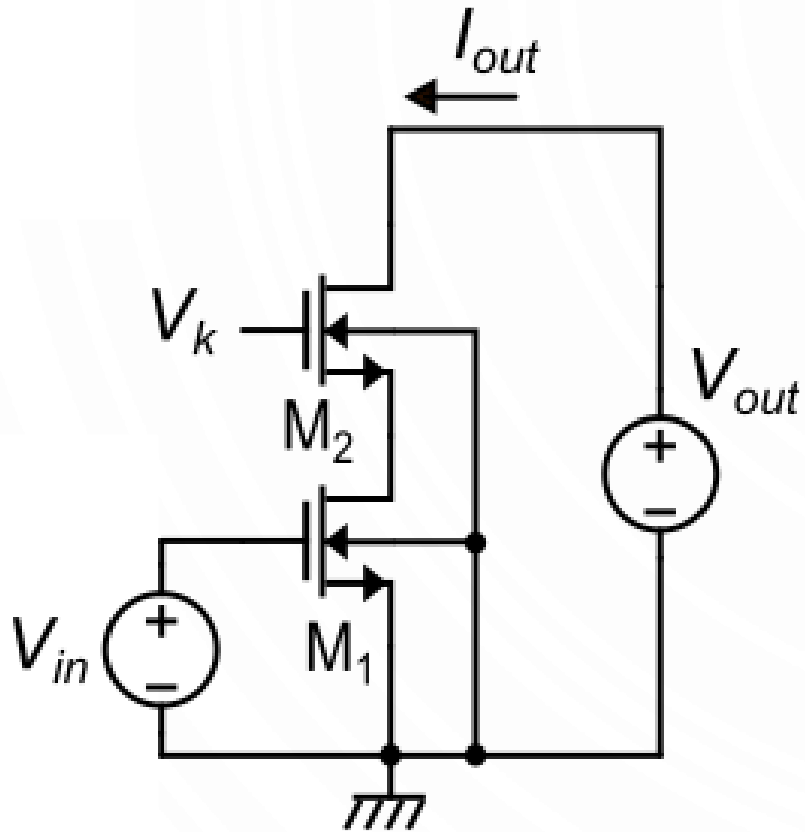
$$R_D I_x + r_{d1} [I_x - g_m V_x] = V_x$$

$$\frac{V_x}{I_x} = \frac{R_D + r_{d1}}{1 + g_m r_{d1}}$$

$$\approx \frac{R_D + r_{d1}}{1 + g_m r_{d1}} = \frac{1}{g_m} \quad \text{if } r_{d1} \gg R_D \text{ (or if } R_D = 0) \text{ and } g_m r_{d1} \gg 1$$



THE CASCODE CONFIGURATION (CASCODE STAGE)



Operating point:

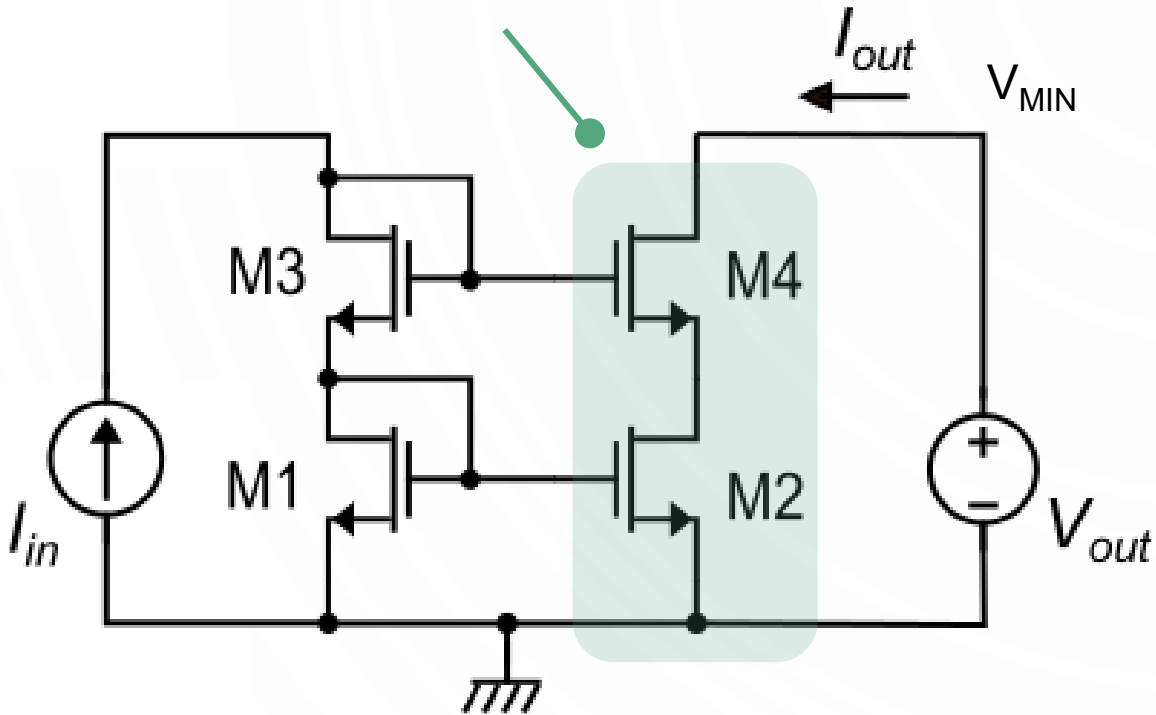
$$V_{DS1} = V_K - V_{GS2}$$

M1: Common source (CS)

M2: Common gate (CG)

STANDARD CASCODE CURRENT MIRROR: PARAMETERS

Cascode structure



Cascode mirror

$$V_{in} = V_{GS1} + V_{GS3} \approx 2V_{GS}$$

$$V_{MIN} = V_{G3,4} - V_T$$

$$V_{MIN} = V_{GS3} + V_{GS1} - V_T$$

$$V_{MIN} = (V_{GS1} - V_T) + (V_{GS3} - V_T) + V_T$$

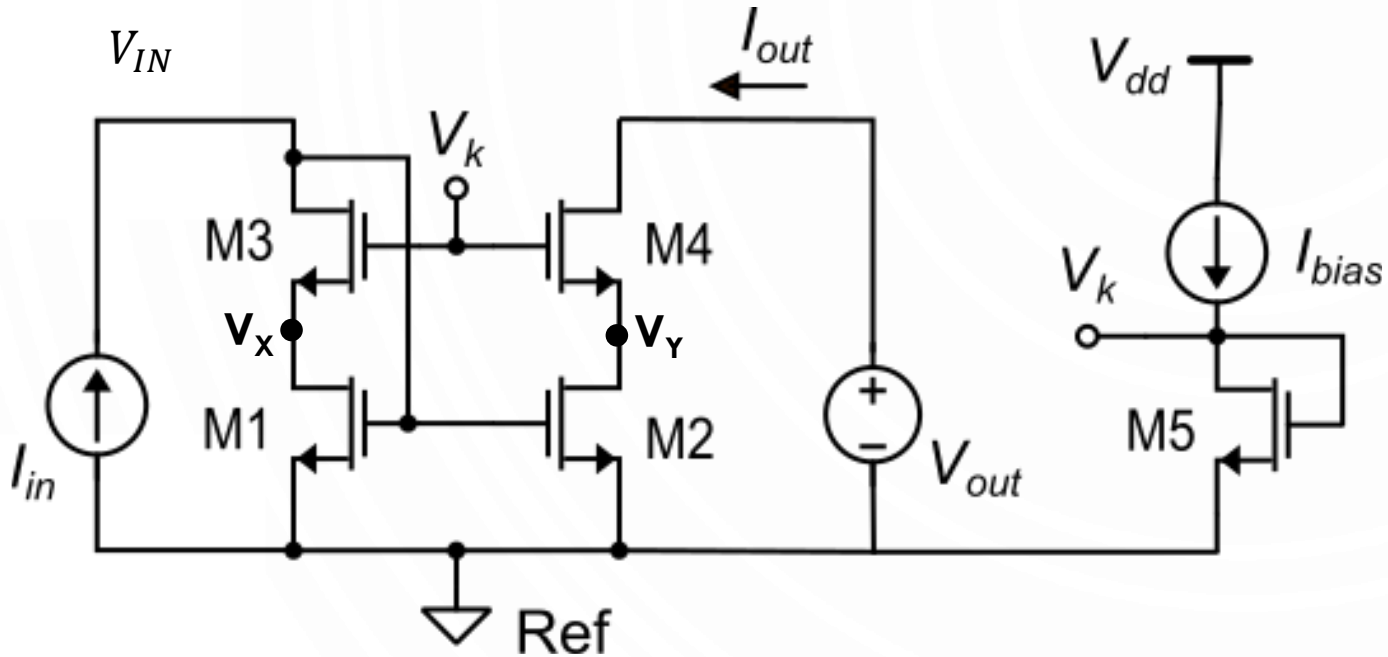
Minimum possible voltage at the output:

$$V_{MIN} \approx V_{GS} + V_{DSAT}$$

Output resistance:

$$R_{out} \cong r_{d4}(g_{m4}r_{d2}) = \frac{1}{\lambda_4 I_{out}}(g_{m4}r_{d2})$$

Bias of a wide-swing cascode structure



If V_K is set as follows:

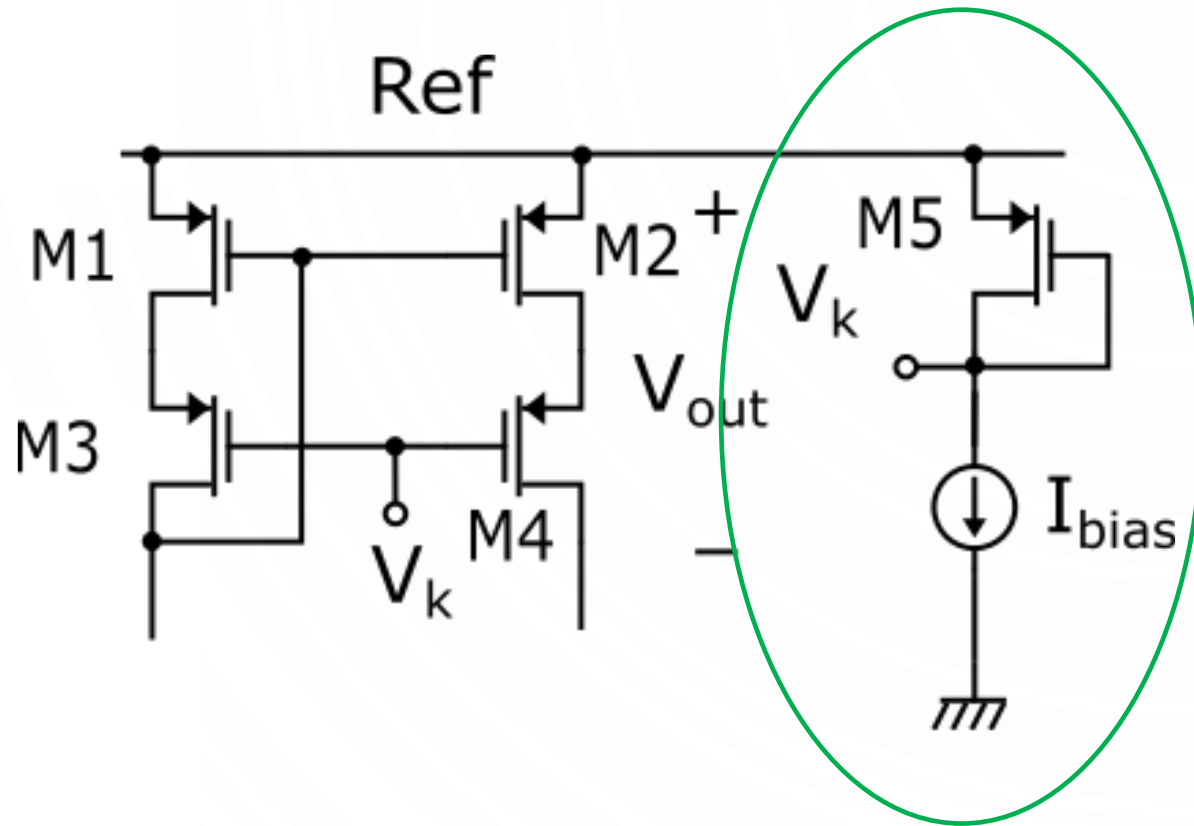
$$V_K = V_t + 2V_{DSAT4}$$

$$V_X = V_Y = V_K - V_{GS3,4} = V_{GS3,4} - V_t = V_{DSAT3,4}$$

The minimum possible output voltage would then be (all transistors identical):

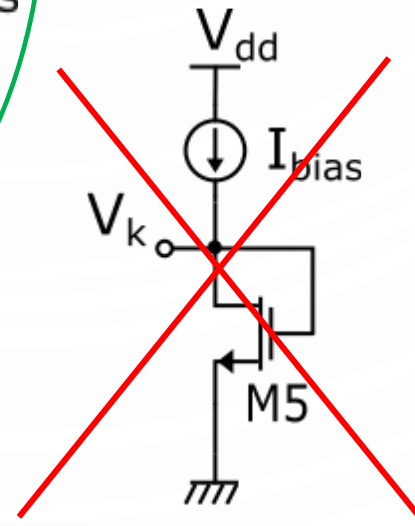
$$V_{OUT} = 2(V_{GS} - V_t) = 2V_{DSAT2,4}$$

P-VERSION OF THE PRECISION WIDE SWING CASCODE MIRROR



Very important rule:

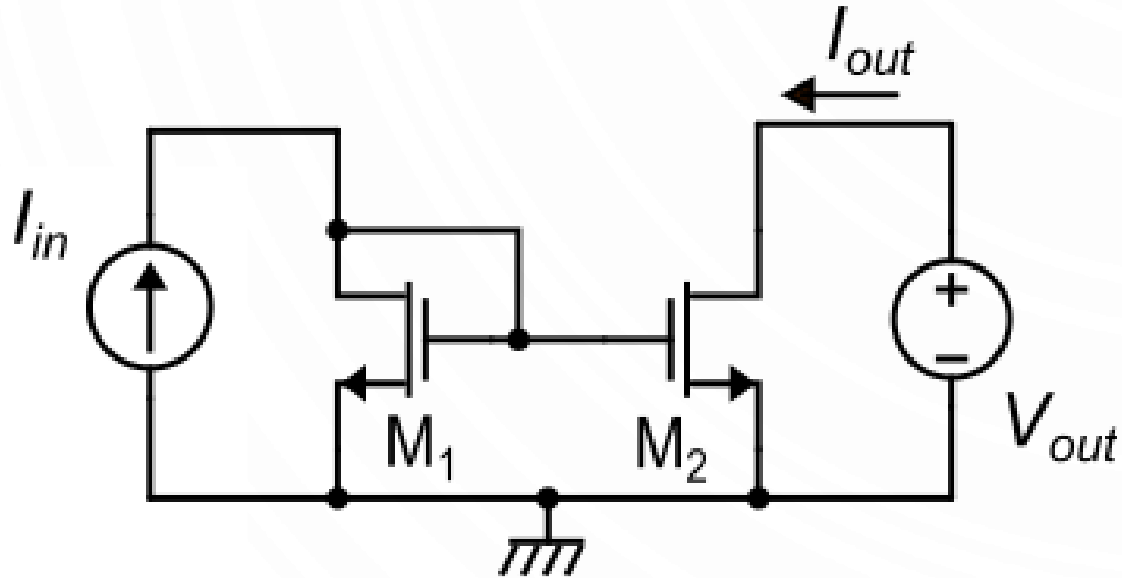
- Voltage V_k must be invariant against V_{dd} variations when measured using Ref as the reference voltage. (i.e. V_k should be referred to node Ref.) The circuit shown in the picture satisfies this requirement.



EFFECT OF MATCHING ERRORS IN CURRENT MIRRORS

CMOS simple current mirror
with $k_M \equiv 1$

M1 is nominally identical to M2



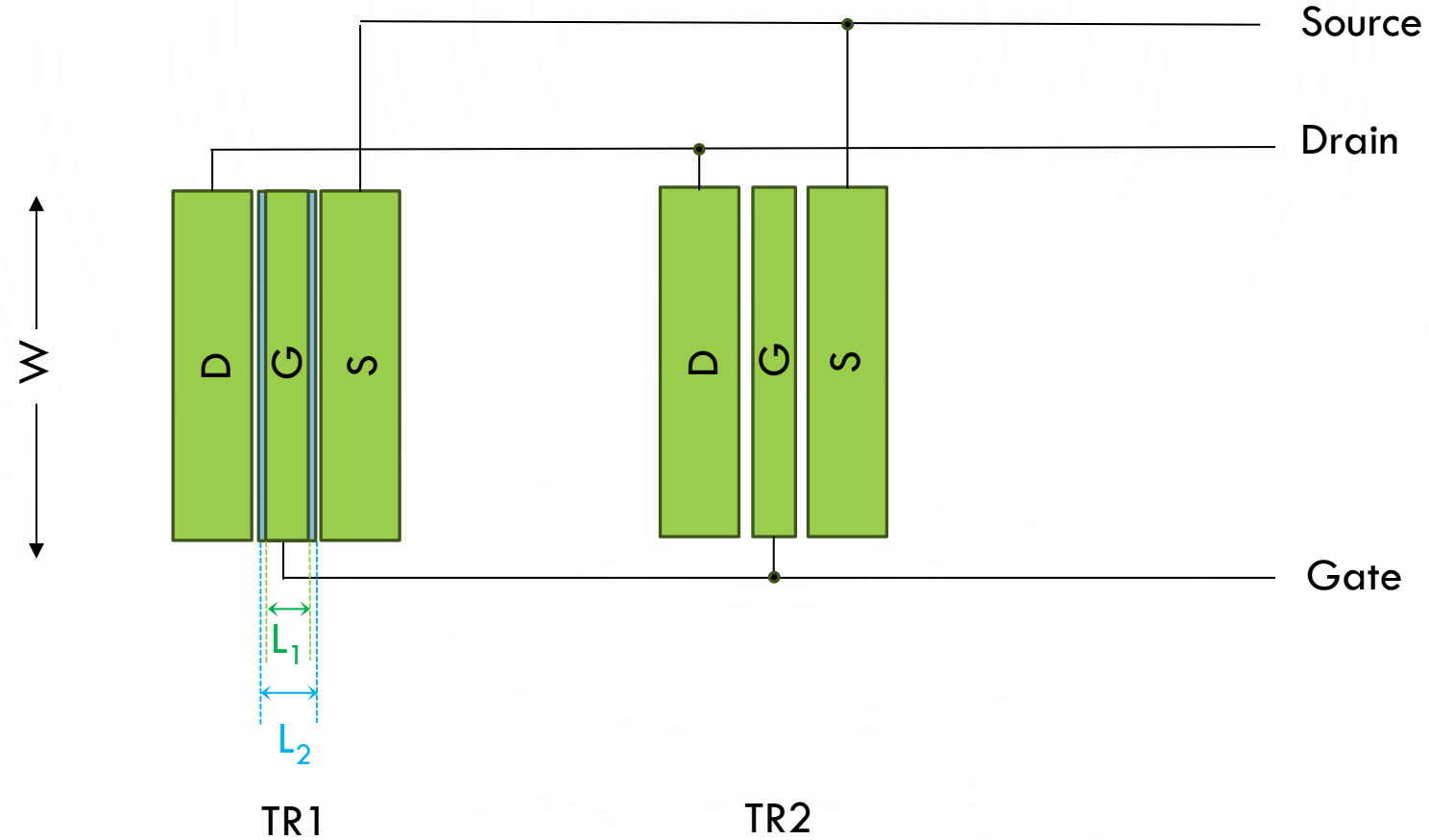
$$\Delta I = I_{out} - I_{in} = I_{D2} - I_{D1} = \Delta I_D$$

$$I_D \cong \frac{\beta}{2} (V_{GS} - V_t)^2$$

$$\beta = \frac{W \mu_n C_{ox}}{L}$$

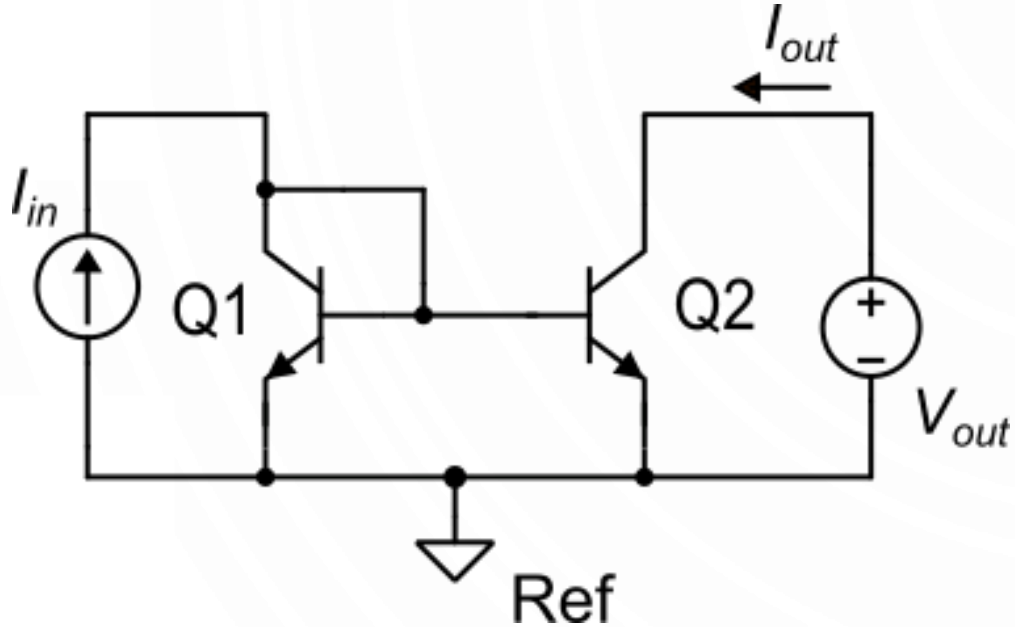
Mismatch between the transistors can occur due to process differences/mismatch (β , V_t).

EFFECT OF MATCHING ERRORS IN CURRENT MIRRORS



TR1 and TR2 shall be as close as possible!

MATCHING ERRORS IN BJT CURRENT MIRRORS



$$I_C \cong I_S e^{\frac{V_{BE}}{V_T}} \quad \Rightarrow \quad V_{BE} = V_T \ln \left(\frac{I_C}{I_S} \right)$$

$$\frac{\Delta I_C}{I_C} = \frac{I_{C2} - I_{C1}}{I_{C1}}$$

$$V_T = \frac{kT}{q}$$

Mismatch depends on temperature and the saturation current I_S (saturation current depends also on the temperature). The collector current has an exponential dependence on V_{BE} and V_T . Paralleling of BJTs must be done with great care and in most cases must be avoided!

References

1) www.halbleiter.org

The Principle of Semiconductor, Tokyo Electron Ltd.

2) Paul Gray, Robert Meyer, "Analysis and Design of Analog Integrated Circuits", John Wiley and Sons

3) Phillip E. Allen, Douglas R. Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2002.